

EXERCISE 02 : SQL FOUNDATIONS
SQL Aggregate Function & Operators

01 The students Table

01. SELECT DISTINCT department
FROM students;

students

department
IT
HR
Finance

02. SELECT department, AVG (age) AS avg-age
FROM students
GROUP BY department;

Students

department	avg-age
IT	20.5
HR	22
Finance	23.3

03. SELECT department, COUNT (*) AS student_count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;

Students

department	Student_count
IT	2
HR	2

04 SELECT student_id, name, age, department
 FROM students
 WHERE age BETWEEN 21 AND 23;

students

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05 SELECT student_id, name, age, department
 FROM students
 WHERE department IN ('IT', 'HR')
 AND age > 21;

Students

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

02 The course Table

06 SELECT department, SUM(credits) AS total_credits
 FROM courses
 GROUP BY department
 HAVING SUM(credits) > 5;

Courses

department	total_credits
IT	11

```

07 SELECT course_id, course_name, department, credits
FROM courses
WHERE credits < 4;

```

Courses

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

```

08 SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

```

Courses

course_id	Course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Q1

03 The enrollments Table

```
09 SELECT MAX (grade) AS max_grade,  
MIN (grade) AS min_grade,  
AVG (grade) AS avg_grade  
FROM enrollments;
```

enrollments

max_grade	min_grade	avg_grade
90	78	84.6

```
10. SELECT course_id, COUNT (*) AS enrollment_count  
FROM enrollments  
GROUP BY course_id
```

enrollments

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

04 The salaries Table

11. SELECT department,
SUM (salary) AS total_salary,
SUM (bonus) AS total_bonus
FROM salaries
GROUP BY department;

Salaries

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12 SELECT department, AVG (salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG (salary) > 55000;

Salaries

department	avg_salary
IT	61000
Finance	70000

13. SELECT employee_id, name, Salary, bonus, (Salary + bonus) AS
total_compensation
FROM Salaries
WHERE salary + bonus > 60000;

Salaries

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

05 The projects Table

14. SELECT department, SUM (budget) AS total_budget,
AVG (budget) AS avg-budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;

projects

department	total_budget	avg total - budget
IT	270000	135000
Finance	80000	80000

15. SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000 AND department
<> 'Marketing';

project

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000